**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Semester Examination – January 2023**

**Course: B. Tech.** **Branch : CE / CSE / CS** **Semester : IV**

**Subject Code & Name: Operating Systems [BTCOC403]**

**Max Marks: 60** **Date:** **Duration: 3 Hrs.**

***Instructions to the Students:***

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

(Level/ CO) Marks

**Q.1 Solve the following questions.** **[12]**

**A)** Explain the storage system hierarchy of operating system with neat diagram. 2

**B)** Write a Short Note on the following: **2**

a) Real-time Embedded System

b) Advantages of Multiprocessor System

**Q.2 Attempt the following questions.** **[12]**

**A)** Describe the actions taken by a kernel to context-switch between processes. **2**

**B)** Describe Process Control Block with suitable Example. **1**

**C)** Determine the average waiting time and draw a Gantt Chart for following process with burst time using Shortest-Job-First scheduling algorithm. **3**

| Process | Burst Time |
|---|---|
| P1 | 6 |
| P2 | 8 |
| P3 | 7 |
| P4 | 3 |

**Q.3 Solve Any Two of the following.** **[12]**

**A)** Discuss the Peterson's solution for the critical-section problem. **2**

**B)** Explain the Dining Philosopher's problem with the structure of philosophers. **2**

**C)** Describe the three requirements to satisfy as a solution to critical-section problem. **1**

**Q.4 Solve any TWO questions of the following.** **[12]**

**A)** Consider a logical address space of 64 pages of 1,024 words each, mapped onto a physical memory of 32 frames. **3**

a) How many bits are there in the logical address?

b) How many bits are there in the physical address?

**B)** Given five memory partitions of 100KB, 500KB, 200KB, 300KB, and 600KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of 212KB, 417KB, 112KB, and 426KB (in order)? Which algorithm makes the most efficient use of memory? **3**

**C)** Consider the following page reference string: **3**

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

How many page faults would occur for the following replacement algorithms, assuming five frames? Remember that all frames are initially empty.

a) LRU replacement

b) FIFO replacement

c) Optimal replacement

**Q.5 Solve Any Two of the following.** **[12]**

**A)** Enlist and Explain in details the various operations performed on the file. **2**

**B)** Describe the following file types with respect to extension used for the file and functioning of the respective file type. **1**

a) Executable
b) object
c) batch
d) library
e) archive
f) source code

**C)** Write the name of the terminology used for the boot-control block and volume-control block in Unix and NT File System. **3**

Consider a file system that uses inodes to represent files. Disk blocks are 8 KB in size, and a pointer to a disk block requires 4 bytes. This file system has 12 direct disk blocks, as well as single, double, and triple indirect disk blocks. What is the maximum size of a file that can be stored in this file system?

***** End *****